SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI

AIM : Association Rule Mining
Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
Generate association rules from the frequent itemsets and calculate their support and confidence.
Interpret and analyze the discovered association rules.

CODE:
```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

# Load dataset
data = pd.read_csv("basket_analysis.csv")

print("=" * 60)
print("DATASET INFORMATION")
print("=" * 60)
print("Dataset Shape:", data.shape)
print("\nFirst 5 Transactions:")
print(data.head())

# --------------------------------------------------
# Convert transaction data to Boolean format
# --------------------------------------------------

data = data.astype(bool)

# --------------------------------------------------
# Find Frequent Itemsets using Apriori
# --------------------------------------------------

frequent_itemsets = apriori(
    data,
    min_support=0.05,
    use_colnames=True
)

frequent_itemsets = frequent_itemsets.sort_values(
```

```
    by="support",
    ascending=False
)

print("\n" + "=" * 60)
print("FREQUENT ITEMSETS")
print("=" * 60)
print(frequent_itemsets.to_string(index=False))

# --------------------------------------------------
# Generate Association Rules
# --------------------------------------------------

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.30
)

# Select important columns
rules = rules[
    ["antecedents", "consequents", "support",
     "confidence", "lift"]
]

# Sort rules by confidence
rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("\n" + "=" * 60)
print("ASSOCIATION RULES")
print("=" * 60)

if len(rules) > 0:
    print(rules.to_string(index=False))
else:
```

```
    print("No association rules found. Try lowering min_support or 
min_threshold.")

# -----------------------------------------------
# Display Top 10 Rules
# -----------------------------------------------

print("\n" + "=" * 60)
print("TOP 10 ASSOCIATION RULES")
print("=" * 60)

if len(rules) > 0:
    print(rules.head(10).to_string(index=False))

# -----------------------------------------------
# Interpretation of Rules
# -----------------------------------------------

print("\n" + "=" * 60)
print("RULE INTERPRETATION")
print("=" * 60)

if len(rules) > 0:
    for i, row in rules.head(10).iterrows():

        antecedent = ", ".join(list(row["antecedents"]))
        consequent = ", ".join(list(row["consequents"]))

        print(
            f"If a customer buys [{antecedent}], "
            f"they are likely to buy [{consequent}]. "
            f"Support = {row['support']:.3f}, "
            f"Confidence = {row['confidence']:.3f}, "
            f"Lift = {row['lift']:.3f}"
        )
else:
    print("No rules available for interpretation.")
```

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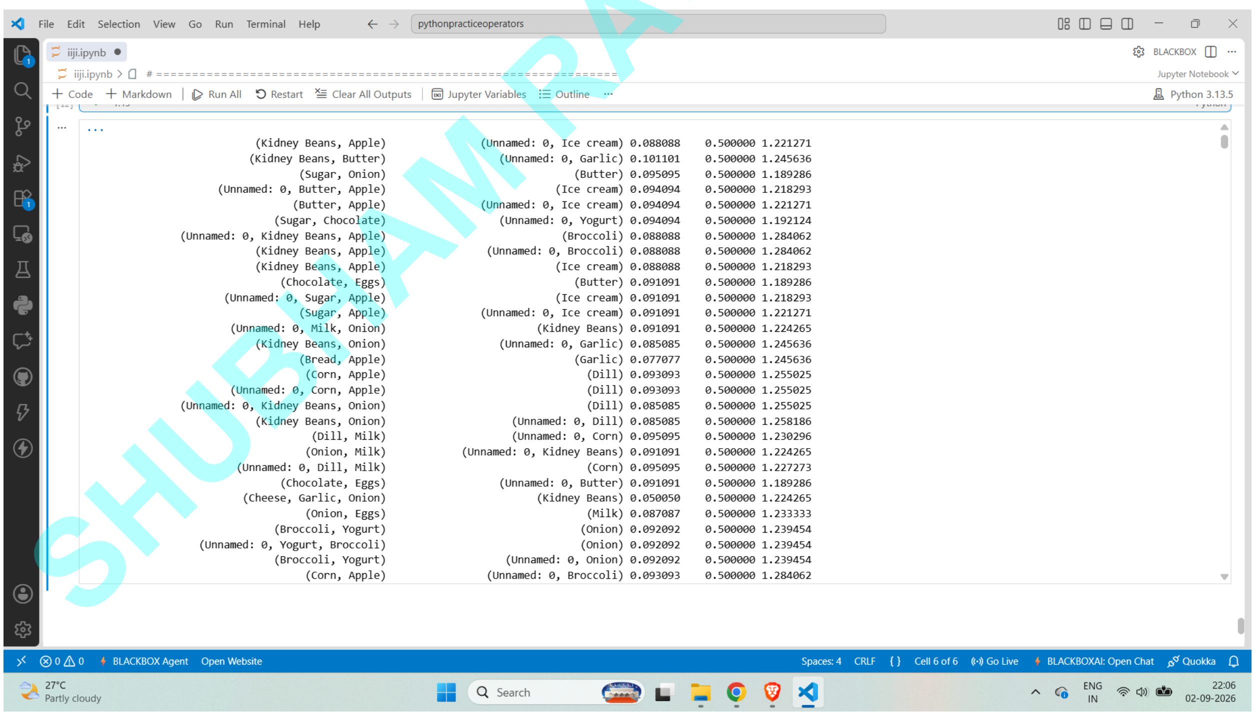
```python
# ----------------------------------------------
# Summary
# ----------------------------------------------

print("\n" + "=" * 60)
print("SUMMARY")
print("=" * 60)
print("Number of Frequent Itemsets:", len(frequent_itemsets))
print("Number of Association Rules:", len(rules))

if len(rules) > 0:
    print("Highest Confidence:",
          round(rules["confidence"].max(), 3))
    print("Highest Lift:",
          round(rules["lift"].max(), 3))
```

OUTPUT



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```
                     (Chocolate, Apple)              (Kidney Beans, Corn) 0.055055   0.300546 1.539723
                           (Corn, Milk)                 (Dill, Chocolate) 0.058058   0.300518 1.508631
                           (Corn, Milk)      (Unnamed: 0, Dill, Chocolate) 0.058058   0.300518 1.516251
                (Unnamed: 0, Corn, Milk)                (Dill, Chocolate) 0.058058   0.300518 1.508631
       (Unnamed: 0, Kidney Beans, Onion)                   (Cheese, Corn) 0.051051   0.300000 1.646703
                   (Kidney Beans, Onion)       (Unnamed: 0, Cheese, Corn) 0.051051   0.300000 1.646703
                   (Kidney Beans, Onion)                   (Cheese, Corn) 0.051051   0.300000 1.646703
                          (Sugar, Eggs)         (Unnamed: 0, Corn, Apple) 0.051051   0.300000 1.611290
              (Unnamed: 0, Sugar, Eggs)                     (Corn, Apple) 0.051051   0.300000 1.611290
                          (Sugar, Eggs)                     (Corn, Apple) 0.051051   0.300000 1.611290

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TOP 10 ASSOCIATION RULES
============================================================
                           antecedents  consequents  support  confidence     lift
        (Kidney Beans, Cheese, Yogurt) (Unnamed: 0) 0.095095         1.0 1.001002
                (Sugar, Butter, Onion) (Unnamed: 0) 0.095095         1.0 1.001002
                                (Milk) (Unnamed: 0) 0.405405         1.0 1.001002
                              (Cheese) (Unnamed: 0) 0.404404         1.0 1.001002
                               (Onion) (Unnamed: 0) 0.403403         1.0 1.001002
                              (Garlic) (Unnamed: 0) 0.401401         1.0 1.001002
      (Onion, Chocolate, Garlic, Milk) (Unnamed: 0) 0.050050         1.0 1.001002
(Kidney Beans, Cheese, Garlic, Yogurt) (Unnamed: 0) 0.050050         1.0 1.001002
  (Broccoli, Cheese, Chocolate, Onion) (Unnamed: 0) 0.050050         1.0 1.001002
             (Garlic, Ice cream, Milk) (Unnamed: 0) 0.081081         1.0 1.001002

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RULE INTERPRETATION
```

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```
                              (Garlic) (Unnamed: 0) 0.401401         1.0 1.001002
      (Onion, Chocolate, Garlic, Milk) (Unnamed: 0) 0.050050         1.0 1.001002
(Kidney Beans, Cheese, Garlic, Yogurt) (Unnamed: 0) 0.050050         1.0 1.001002
  (Broccoli, Cheese, Chocolate, Onion) (Unnamed: 0) 0.050050         1.0 1.001002
             (Garlic, Ice cream, Milk) (Unnamed: 0) 0.081081         1.0 1.001002

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RULE INTERPRETATION
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If a customer buys [Kidney Beans, Cheese, Yogurt], they are likely to buy [Unnamed: 0]. Support = 0.095, Confidence = 1.000, Lift = 1.001
If a customer buys [Sugar, Butter, Onion], they are likely to buy [Unnamed: 0]. Support = 0.095, Confidence = 1.000, Lift = 1.001
If a customer buys [Milk], they are likely to buy [Unnamed: 0]. Support = 0.405, Confidence = 1.000, Lift = 1.001
If a customer buys [Cheese], they are likely to buy [Unnamed: 0]. Support = 0.404, Confidence = 1.000, Lift = 1.001
If a customer buys [Onion], they are likely to buy [Unnamed: 0]. Support = 0.403, Confidence = 1.000, Lift = 1.001
If a customer buys [Garlic], they are likely to buy [Unnamed: 0]. Support = 0.401, Confidence = 1.000, Lift = 1.001
If a customer buys [Onion, Chocolate, Garlic, Milk], they are likely to buy [Unnamed: 0]. Support = 0.050, Confidence = 1.000, Lift = 1.001
If a customer buys [Kidney Beans, Cheese, Garlic, Yogurt], they are likely to buy [Unnamed: 0]. Support = 0.050, Confidence = 1.000, Lift = 1.001
If a customer buys [Broccoli, Cheese, Chocolate, Onion], they are likely to buy [Unnamed: 0]. Support = 0.050, Confidence = 1.000, Lift = 1.001
If a customer buys [Garlic, Ice cream, Milk], they are likely to buy [Unnamed: 0]. Support = 0.081, Confidence = 1.000, Lift = 1.001

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SUMMARY
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Number of Frequent Itemsets: 1892
Number of Association Rules: 10187
Highest Confidence: 1.0
Highest Lift: 1.826

Association Rule Mining using Apriori completed successfully.
```

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